

IMPACT ANALYSIS

CR-010: Student Profile Image Upload for the Admin Student Management

1.Change Summary

CR-002 introduces a student profile image upload facility to the administrator's Add Student functionality.

Administrators will be able to select and upload a profile image when creating a new student. The system will support **PNG, JPG, and JPEG** image formats. The uploaded image will be associated with the correct student record and displayed in the student list or student details section.

The change is intended to improve student identification and the usability of the Student Management System.

2.Impact Assessment

Area	Impact	Assessment
Frontend	Moderate	Add Student form will be updated with a profile image upload field and image validation.
Backend	Moderate	Backend upload handling and file validation will be required to securely process uploaded images.
API	Low - Moderate	Existing student creation functionality may need to support multipart/form-data for image upload.
Database	Low	A profile image path/reference may need to be stored with the student record.
Authentication	None	Existing administrator authentication remains unchanged.
Authorization	Low	Existing administrator authorization must continue to restrict student image uploads to authorized users.
Performance	Low	Small profile images will have limited performance impact when properly validated and stored.
CRUD	Low	Existing Add, Edit and Delete operations must continue to work correctly after the image feature is added.
Storage	Moderate	Uploaded images require a suitable server-side upload directory or storage location.
Deployment	Moderate	The upload directory and image-serving configuration must be available in the deployed application.
Security	Moderate	File type validation and secure upload handling are required to prevent unsafe files.

3.Frontend Impact

The main affected area is the administrator's **Add Student** interface.

Potential changes include:

- Profile image upload field.
- File selection control.
- Accepted file format validation.
- PNG, JPG and JPEG support.
- Invalid file error message.
- Optional image preview.
- Profile image display in the student list.
- Profile image display in the student details section.
- Placeholder image when no image is available.
- Responsive styling for the upload interface.

Add Student

Registration Number: [ST001]

First Name: [Kamal]

Last Name: [Perera]

Email: [kamal@gmail.com]

Phone: [077xxxxxxx]

Profile Image:
[Choose File]

Supported formats: PNG, JPG, JPEG

[Add Student]

The implementation must preserve the existing student form fields and CRUD functionality.

4.Backend, API and Database Impact

Unlike CR-001, this change requires some backend support because an actual image file must be uploaded and stored.

The backend may require:

- Multipart form-data handling.
- Image file validation.
- Upload directory configuration.
- Secure file naming.
- Image path generation.
- Association of the image path with the correct student record.
- Serving uploaded images to the frontend.

The existing student creation functionality should continue to work.

The database may require an additional field such as:
`profile_photo`

The database should store the **image path/reference**, rather than storing the complete image file as binary data.

Example:

`/uploads/students/student_001.jpg`

No new student-related table is expected to be required.

5.Security Impact

The change has a moderate security impact because the system will accept files from an administrator's computer.

Existing authentication and authorization mechanisms must remain unchanged.

The image upload feature must:

- Allow only authorized administrators to upload images.
- Accept only PNG, JPG and JPEG formats.
- Reject unsupported file types.
- Validate the uploaded file on the server.
- Avoid allowing executable files to be uploaded as images.
- Use safe file names rather than directly trusting user-provided file names.
- Avoid unsafe HTML rendering.
- Prevent uploaded files from being treated as executable server-side code.
- Avoid exposing additional sensitive student information.
- Preserve the existing administrator authorization controls.

Examples of files that must be rejected:

`student.pdf`
`student.exe`
`student.js`
`student.php`
`student.html`
`student.txt`

Supported files:

`student.png`
`student.jpg`
`student.jpeg`

6.CRUD and Functional Impact

Existing administrator CRUD operations must continue to work after the profile image feature is introduced.

Particular attention is required for:

- Adding a student with an image.
- Adding a student without an image.
- Displaying the correct image for the correct student.
- Editing an existing student.
- Deleting an existing student.
- Maintaining the correct `student_id`.
- Preserving existing student information.

The image upload feature must not change the identity of a student record.

For example:

```
Student ID: ST001  
Name: Kamal Perera  
Image: kamal.jpg
```

The image must remain associated with:

```
student_id = ST001
```

and must not accidentally be assigned to another student.

7.Performance Impact

The expected performance impact is **low to moderate**.

Profile images will require additional storage and network transfer when they are uploaded and displayed.

To reduce unnecessary performance impact:

- Only supported image formats should be accepted.
- Reasonable file-size limits should be applied.
- Images should be stored as files rather than large binary values in the database.
- Only the required image should be loaded for each student.
- A suitable placeholder should be used when no image exists.

For the current expected student population, the performance impact should remain manageable.

If the number of students and uploaded images becomes very large, image compression, resizing, dedicated storage, or CDN-based image delivery may be considered in a future change.

8. Testing Impact

Manual and functional testing should cover:

- Add student without an image.
- Upload PNG image.
- Upload JPG image.
- Upload JPEG image.
- Reject PDF file.
- Reject EXE file.
- Reject TXT file.
- Reject other unsupported file formats.
- Display invalid file validation message.
- Successful image upload.
- Correct image displayed for the correct student.
- Student without image displays placeholder.
- Edit student after image upload.
- Delete student after image upload.
- Correct `student_id` maintained.
- Existing student CRUD functionality.
- Administrator authentication.
- Administrator authorization.
- Image display after page refresh.
- Image display after restarting the backend, where applicable.

9. Risk Analysis

Risk	Impact	Mitigation
Unsafe file uploaded	High	Validate file type on both frontend and backend.
Unsupported image format uploaded	Medium	Allow only PNG, JPG and JPEG.
Wrong image associated with student	High	Associate image using the correct <code>student_id</code> .
Existing Add Student functionality breaks	High	Perform regression testing after implementation.
Invalid file name causes security issue	High	Generate safe server-side file names.
Very large image affects performance	Medium	Apply a reasonable file-size limit.
Student has no image	Low	Display a default placeholder image.
Image path becomes invalid	Medium	Store and generate consistent upload paths.
Upload directory unavailable after deployment	Medium	Configure and verify the upload directory during deployment.

Unauthorized user uploads image	High	Preserve existing administrator authentication and authorization.
Too many images consume storage	Medium	Apply file-size limits and storage management.

10. Overall Impact and Recommendation

Overall Impact: Moderate

CR-002 is primarily a **frontend and backend enhancement** that introduces profile image upload functionality to the Admin Student Management module.

The change requires modifications to the Add Student interface, backend file-upload handling, image validation, and potentially the student database structure to store the image path/reference.

The existing authentication, authorization, and student CRUD functionality should remain unchanged.

The change is technically feasible and recommended for implementation because it improves student identification and provides a more user-friendly student management experience.

Key Implementation Considerations

The most important considerations are:

1. Only **PNG, JPG and JPEG** files should be accepted.
2. File validation should be performed on both the frontend and backend.
3. Uploaded files should be stored securely.
4. The database should store the image path/reference rather than the complete image.
5. The image must always be associated with the correct `student_id`.
6. Existing Add, Edit and Delete functionality must continue to work.
7. Existing administrator authentication and authorization must remain unchanged.
8. A placeholder should be displayed when a student has no profile image.

Recommendation: Proceed with implementation after formal approval, followed by functional, security, CRUD regression, and deployment testing.